nature portfolio

Peer Review File

A Multimodal Framework for Comprehensive Driver Variant Prediction in Cancer

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This file contains all reviewer reports order by version, followed by all author rebuttals order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
I want to thank the authors for revising their manuscript "ModVAR: A Multimodal Framework for Comprehensive Driver Variant Prediction Cancer". The revised manuscript is vastly improved and should be accepted for publication. My comments and suggestions were adequately addressed.

Reviewer #2

(Remarks to the Author)
this study, the authors present ModVAR, a multimodal model for identifying driver variants. ModVAR integrates features from three distinct modalities: sequences, protein tertiary structures, and omics data. ModVAR demonstrates superior or comparable performance to existing methods predicting experimentally validated and clinically relevant variants across multiple benchmark datasets. The authors have sufficiently addressed the comments from the previous round of revision. We are providing several additional comments that should be addressed or clarified before the manuscript is suitable for publication.

Comments:
1. The authors used the number of variants corresponding to specific variant types (e.g., frame shift insertion/deletion, missense mutation) as features at both the gene level (Table S5) and variant level (Table S7). Could the authors clarify how these counts were aggregated for gene-level features? Given that variant types are inherently associated with individual variants, did the inclusion of gene-level features significantly contribute to model performance? Additionally, there appear to be redundant features within Table S5, for example, 'Nonsense_Mutation' and 'nonsense', or 'Splice_Site' and 'splice site'. Do these terms have different definitions?
2. incorporating cancer-specific statistical annotations (Table S7), the authors state that they used a 100 bp window surrounding each variant. However, given that even closely located coding variants can have distinct functional impacts, aggregating information across such a window may introduce bias from nearby variants. Did the authors consider applying a distance-based weighting to better account for the proximity of annotations relative to the query variant when aggregating within the window?

Version 1:

Reviewer comments:

Reviewer #3

(Remarks to the Author)
I would like to thank the authors for revising their manuscript response to my comments. They have sufficiently addressed all of my concerns.

Responses to the reviewers' comments:

1. Reply to the Reviewer #1

Comment 1:

"I want to thank the authors for revising their manuscript "ModVAR: A Multimodal Framework for Comprehensive Driver Variant Prediction in Cancer". The revised manuscript is vastly improved and should be accepted for publication. My comments and suggestions were adequately addressed.".

Reply 1:

We sincerely appreciate your positive acknowledgment of our previous revisions. We are also truly grateful for the constructive feedback you provided earlier, as it has helped us improve the manuscript from a more comprehensive perspective. Your valuable suggestions have been instrumental in refining our work.

2. Reply to the Reviewer #2

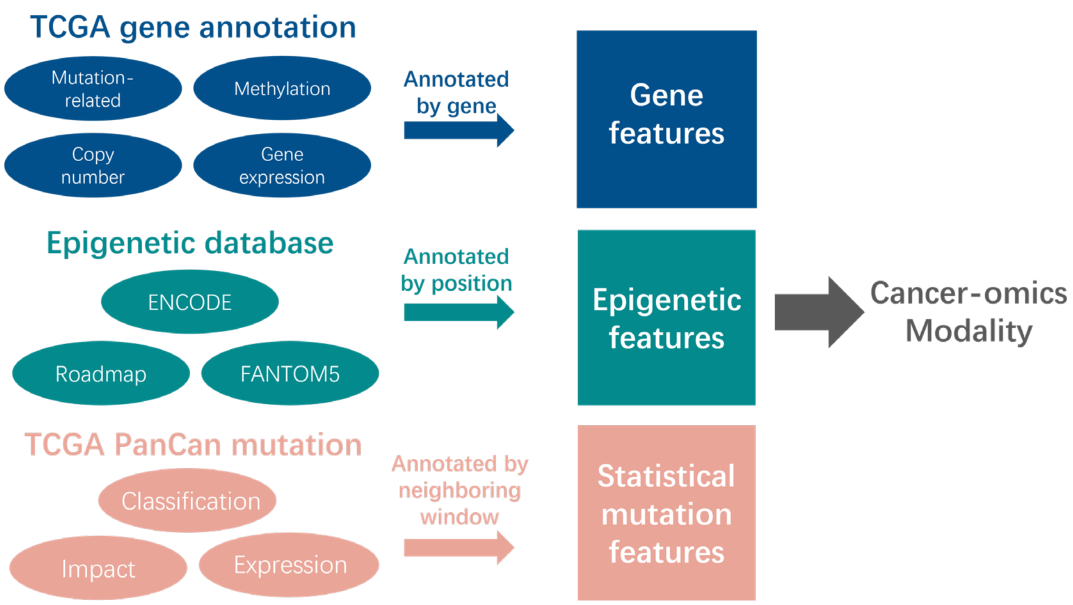
Comment 1:

"1. The authors used the number of variants corresponding to specific variant types (e.g., frame shift insertion/deletion, missense mutation) as features at both the gene level (Table S5) and variant level (Table S7). Could the authors clarify how these counts were aggregated for gene-level features? Given that variant types are inherently associated with individual variants, did the inclusion of gene-level features significantly contribute to model performance? Additionally, there appear to be redundant features within Table S5, for example, 'Nonsense_Mutation' and 'nonsense', or 'Splice_Site' and 'splice site'. Do these terms have different definitions?"

Reply 1:

First, we would like to sincerely thank you for recognizing the modifications we made in our previous response! As you pointed out, the information related to gene features is indeed vital. Therefore, in the revised manuscript and supplementary materials, we have provided a more detailed description of the feature processing workflow for this section. **Specifically, we collected DNA methylation, copy number variation (CNV) data, and somatic mutation data across 33 types of cancer from the TCGA Pan-cancer Atlas. To extract CNV features, we matched data based on the 'Tumor Sample Barcode' from mutation profiles and calculated the mean CNV values. We then established the relationship between 'Hugo Symbol' and 'Tumor Sample Barcode' in somatic mutation data, linking protein-coding genes with CNV intervals. The mean and variance of CNVs for each gene were calculated, separately considering upregulation and downregulation. We obtained DNA methylation data from the TCGA Pancancer Atlas, calculated the mean and variance for each gene, and mapped the DNA methylation data to specific genes based on CpG islands and associated genes. Finally, we generated mutation-related features by counting the number of different somatic mutations associated with each gene.**

In order to provide a clearer representation of the workflow for extracting our cancer-omics features, we have included a diagram of the extraction process in the supplementary materials:



Supplementary Fig. 11| The feature extraction workflow of cancer omics modality.

The features of the cancer-omics modality are primarily made up of three components: gene-level features, epigenetic features, and statistical mutation features. These components correspond to three levels: gene, site, and neighboring window. Features are extracted and summarized from various databases and feature categories at each level. (Refer to Supplementary information, Supplementary Fig. 11)

Similarly, we have improved the descriptions in the manuscript, and added more detailed information in Table S5:

" *For gene annotation, we extracted 46 columns of relevant features, including DNA methylation, Copy number variation (CNV) data, and somatic mutation data across 33 types of cancer from the TCGA Pan-cancer Atlas.*" (Refer to Section 4.2,L791-L794)

Feature Name	**Category**	**Description**
Frame_Shift_Del	Mutation-related	Translational effect of variant allele
Frame_Shift_Ins		Translational effect of variant allele
In_Frame_Del		Translational effect of variant allele
In_Frame_Ins		Translational effect of variant allele
Missense_Mutation		Translational effect of variant allele
Nonsense_Mutation		Translational effect of variant allele
Nonstop_Mutation		Translational effect of variant allele
Silent		Translational effect of variant allele
Splice_Site		Translational effect of variant allele
Translation_Start_Site		Translational effect of variant allele
DEL		Type of mutation. DEL (deletion) is deletion-induced mutation
INS		Type of mutation. INS (insertion) is insertion-induced mutation
SNP		Type of mutation. SNP (single-nucleotide polymorphism) is 1consecutive nucleotides
pnum		Count the number of times the gene appears in the patient sample
gene length		Length of Gene
silent fraction		Translational effect of variant allele
nonsense fraction		Translational effect of variant allele
splice site fraction		Translational effect of variant allele
missense fraction		Translational effect of variant allele
recurrent missense fraction		Translational effect of variant allele
normalized missense position entropy		Frequency of missense mutations at each location
frameshift indel fraction		Translational effect of variant allele
inframe indel fraction		Translational effect of variant allele
normalized mutation entropy		Frequency of mutation types at each base location
Mean Missense MGAEntropy		Mean of Missense MGAEntropy

lost start and stop fraction		Translational effect of variant allele
missense to silent		Ratio of missense to silent mutations
non-silent to silent		Ratio of non-silent to silent mutations.
all_mean	Copy number	Mean of copy number variations
all_var		Variance of copy number variations
up_mean		Mean of copy number increase
up_var		Variance of copy number increase
down_mean		Mean of methylation
down_var		Variance of methylation
dnam_mean	Methylation	Mean of methylation
dnam_var		Variance of methylation
expression_CCLE	Gene expression	Average expression of a gene in the Cancer Cell Line Encyclopedia
inactivating p-value	Other	p-value of inactivating single nucleotide variants
entropy p-value		p-value of entropy
vest p-value		p-value of vest score
combined p-value		p-value of VEST scores and missense clustering using Fisher's method.
Mean VEST Score		Mean of VEST Score
replication_time		DNA replication time during cell cycle
HiC_compartment		HiC measures open vs closed chromatin
gene_betweeness		Fraction of shortest paths that pass through a gene's node in the BioGrid interaction network
gene_degree		Number of other genes that are connected in the BioGrid interaction network

Supplementary Table.6| Detailed list of ModVAR gene features in the cancer-omics modality. *To extract ModVAR gene features in the cancer-omics modality, we gathered DNA methylation, CNV, and somatic mutation data across 33 cancer types from the TCGA Pan-cancer Atlas, resulting in a total of 46 dimensions, all organized at the gene level. CNV features were extracted by linking mutation profiles with gene data, calculating mean and variance for each gene. DNA methylation data was mapped to genes based on CpG islands. Mutation-related features were generated by counting somatic mutations for each gene. ModVAR retrieves gene-specific statistical annotation features based on the gene of each mutation to be predicted.* (Refer to Supplementary information, Table. S6)

Additionally, regarding your question, "*There appear to be redundant features within Table S5, for example, 'Nonsense_Mutation' and 'nonsense', or 'Splice_Site' and 'splice site'. Do these terms have different definitions?*", these features are all mutation-related gene features, but they have distinct meanings. The former (uppercase and with underscores) fields are derived from the TCGA annotation tool Oncotator[1], where the values are integers representing the number of mutations of this type within a gene. The latter (lowercase and without underscores) fields come from the 20/20+ tool's feature extraction module[2], with values being decimals representing the fraction of mutations of this type within a gene. Therefore, in this work, we retained all attributes from both categories to provide a more comprehensive set of gene-level features with detailed information. To ensure clearer expression, we have updated the field names in the supplementary materials table (see the Supplementary Table.6 above).

Finally, regarding your question about whether gene-level features significantly contribute to the model's predictions, we conducted an ablation analysis specifically for gene-level features, and the results are shown in Table below. As seen from the table, the presence of gene features provides varying degrees of improvement to the model across different datasets. **The most significant performance gains were observed in the mutation cluster, In vitro, and CIViC datasets, while improvements were also noted in the other datasets. These results demonstrate that gene-level features contribute valuable variant-related information in most scenarios. In certain contexts, such as druggable variants, gene-level features may hold crucial information that is essential for task understanding, which corresponds to the reduced contribution of protein modality in the analysis of this subset.** Due to the varying roles of different genes, we believe that incorporating gene-level omics features is highly beneficial for certain tasks, especially for complex tasks like target identification.

mutation clustering patterns	**AUPRC**	**AUROC**
ModVAR	**0.9361**	**0.9537**
ModVAR_without_gene	0.8702	0.9120
In vitro tumor formation		
ModVAR	**0.5971**	**0.7570**
ModVAR_without_gene	0.5734	0.7414
In vivo cell viability		
ModVAR	**0.8426**	**0.7562**
ModVAR_without_gene	0.8378	0.7555
DisProt		
ModVAR	**0.8753**	**0.9093**
ModVAR_without_gene	0.8658	0.9071
ClinVar Tier1		
ModVAR	**0.9848±0.003**	**0.9849±0.003**
ModVAR_without_gene	0.9837±0.003	0.9839±0.003
CIViC		
ModVAR	**0.8581±0.020**	**0.9165±0.022**
ModVAR_without_gene	0.8268±0.046	0.9019±0.023

Supplementary Table.5| Ablation analysis results of gene features on each benchmark.

Additionally, we have included a discussion in the revised manuscript:

"*Furthermore, we conducted an ablation study on the gene features within the cancer-omics modality (Table.S5) and found that, gene features consistently improved the performance of ModVAR. The most significant improvements were observed in applications such as cancer druggable variants, which corresponds to the reduced contribution of protein modality in the SHAP analysis of these subsets. These analyses demonstrate the advantage of integrating complementary features from different modalities, which enhances the model's ability to predict variant drivers more effectively.*" (Refer to Section Discussion, L669-L676)

Comment 2:

"2. In incorporating cancer-specific statistical annotations (Table S7), the authors state that they used a 100 bp window surrounding each variant. However, given that even closely located coding variants can have distinct functional impacts, aggregating information across such a window may introduce bias from nearby variants. Did the authors consider applying a distance-based weighting to better account for the proximity of annotations relative to the query variant when aggregating within the window?"

Reply 2:

Thank you for your constructive feedback regarding the extraction of cancer-specific statistical annotations. We have carefully considered your comments and agree that incorporating a distance measure for the annotations within the 100bp window is indeed important and worth exploring experimentally. As a result, we have added a distance-weighted metric to this feature extraction process. Specifically, we calculate the distance between each background mutation sample and the annotation point within the 100bp window, dividing this distance by 100 (with the maximum distance being 100) to obtain a distance weight for that statistical point. Then, for each annotation point, we multiply each background mutation feature within the window by its corresponding distance weight and sum them to obtain the cancer-specific statistical annotations with distance metrics. The detailed Equation is as follows:

$$W_i = \min(|start_i - pos|, |end_i - pos|) / \text{window} \tag{1}$$

$$F_m = \sum_i W_i f_m \tag{2}$$

Here, i represents the i^{th} background mutation within the window, m denotes the m^{th} feature in the statistic, *start* and *end* refer to the start and end positions of the background mutation, *pos* represents the annotation point from which features are to be extracted, *window* indicates the window length, W denotes the distance weight, f and F refer to the statistical annotation features.

Using the approach described above, we re-derived the cancer-specific annotations with distance-weighted metrics and conducted comparative experiments with the original method. The specific results are shown in the table below. From the results, we observed that the inclusion of distance metrics did not significantly alter the outcomes in most test sets. On the ClinVar T1 test set, the distance-based feature provided a slight performance improvement. In contrast, most other test sets showed a marginal decrease in performance.

mutation clustering patterns	**AUPRC**	**AUROC**
Original feature	**0.9361**	**0.9537**
Distance based feature	0.9295	0.9491
In vitro tumor formation		
Original feature	**0.5971**	**0.7570**
Distance based feature	0.5814	0.7532
In vivo cell viability		
Original feature	**0.8426**	**0.7562**
Distance based feature	0.8039	0.7162
DisProt		
Original feature	**0.8753**	**0.9093**
Distance based feature	0.8735	0.9090
ClinVar Tier1		

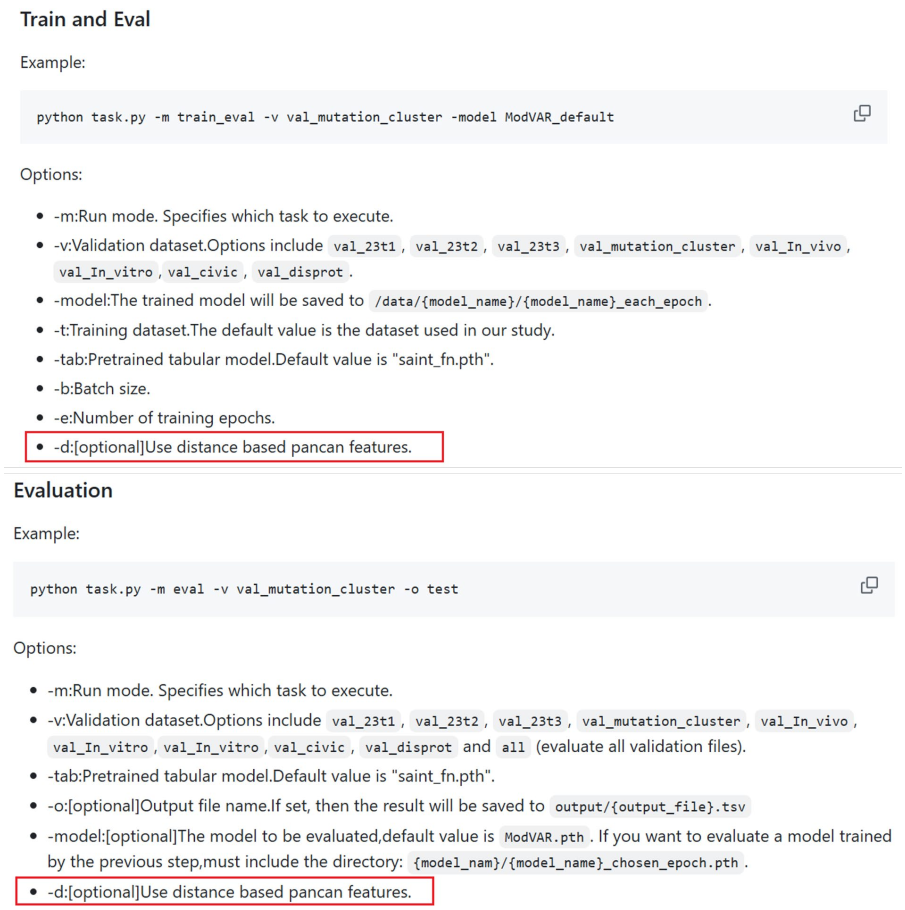
Original feature	0.9848±0.003	0.9849±0.003
Distance based feature	**0.9849±0.003**	**0.9850±0.003**
CIViC		
Original feature	**0.8581±0.020**	**0.9165±0.022**
Distance based feature	0.8495±0.044	0.9141±0.022

Table.R1| Exploration results of distance-based cancer-specific features on each benchmark.

We believe that considering distance metrics is still a worthwhile direction for further exploration. Therefore, in the revised manuscript, the above study and conclusions have been included as part of the future directions and discussed in the last section:

"*One aspect worth considering is a more refined processing of the features for each modality. For example, in the cancer-omics modality, the current strategy for cancer-specific statistical annotations is to use a 100bp window around the mutation site to measure nearby mutations. However, this approach does not take into account the varying influence of mutations based on their distance from the central site. We have already tried designing features that incorporate distance weighting, but the results only showed a slight performance difference. Therefore, in the future, ModVAR's feature extraction module can be optimized to achieve more precise modeling results.*" (refer to Section Discussion, L698-L706)

Additionally, we have provided a selectable switch in the relevant code on GitHub (refer to task.py, readme.md), allowing users to choose whether to run the program with the additional distance-weighted features based on their needs.



Train and Eval

Example:

```
python task.py -m train_eval -v val_mutation_cluster -model ModVAR_default
```

Options:

- -m:Run mode. Specifies which task to execute.
- -v:Validation dataset.Options include `val_23t1`, `val_23t2`, `val_23t3`, `val_mutation_cluster`, `val_In_vivo`, `val_In_vitro`, `val_civic`, `val_disprot`.
- -model:The trained model will be saved to `/data/{model_name}/{model_name}_each_epoch`.
- -t:Training dataset.The default value is the dataset used in our study.
- -tab:Pretrained tabular model.Default value is "saint_fn.pth".
- -b:Batch size.
- -e:Number of training epochs.
- -d:[optional]Use distance based pancan features.

Evaluation

Example:

```
python task.py -m eval -v val_mutation_cluster -o test
```

Options:

- -m:Run mode. Specifies which task to execute.
- -v:Validation dataset.Options include `val_23t1`, `val_23t2`, `val_23t3`, `val_mutation_cluster`, `val_In_vivo`, `val_In_vitro`, `val_In_vitro`, `val_civic`, `val_disprot` and `all` (evaluate all validation files).
- -tab:Pretrained tabular model.Default value is "saint_fn.pth".
- -o:[optional]Output file name.If set, then the result will be saved to `output/{output_file}.tsv`
- -model:[optional]The model to be evaluated,default value is `ModVAR.pth`. If you want to evaluate a model trained by the previous step,must include the directory: `{model_nam}/{model_name}_chosen_epoch.pth`.
- -d:[optional]Use distance based pancan features.

Fig. R1| The updated functions on Github.

Reference:

1. Ramos, A. H. et al. Oncotator: Cancer Variant Annotation Tool. Human Mutation 36, E2423–E2429 (2015).
2. Tokheim, C. J., Papadopoulos, N., Kinzler, K. W., Vogelstein, B. & Karchin, R. Evaluating the evaluation of cancer driver genes. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 14330–14335 (2016).